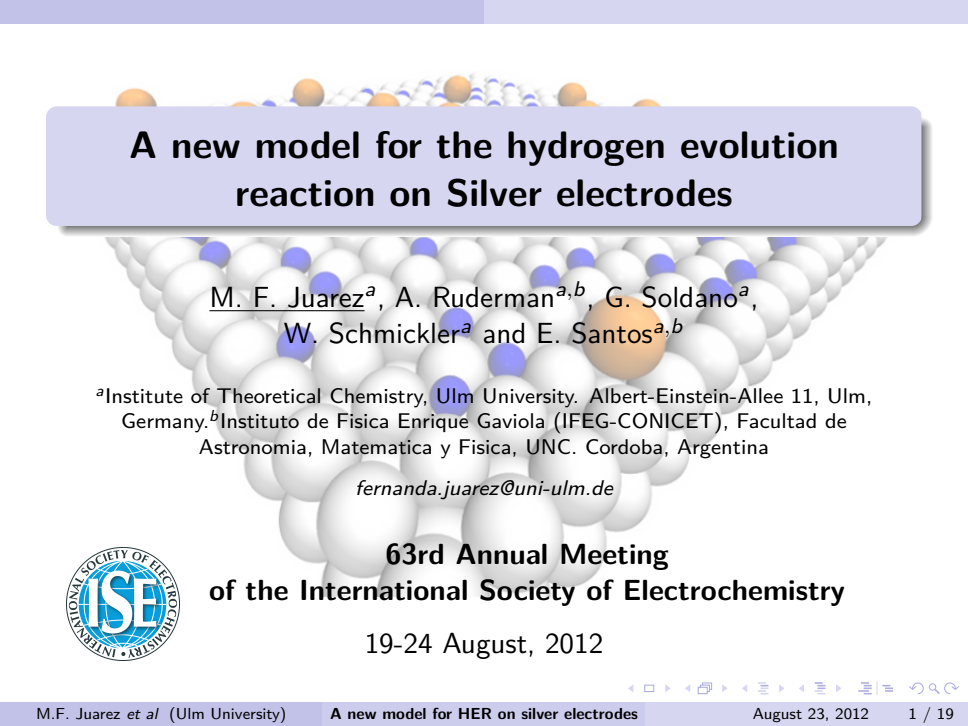




A new model for the hydrogen evolution reaction on Silver electrodes

M. F. Juárez^a, A. Ruderman^{a,b}, G. Soldano^a,
W. Schmickler^a and E. Santos^{a,b}

^aInstitute of Theoretical Chemistry, Ulm University. Albert-Einstein-Allee 11, Ulm, Germany.
^bInstituto de Física Enrique Gaviola (IFEG-CONICET), Facultad de Astronomía, Matemática y Física, UNC. Córdoba, Argentina

fernanda.juarez@uni-ulm.de



**63rd Annual Meeting
of the International Society of Electrochemistry**

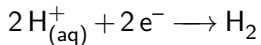
19-24 August, 2012

Motivation



HER reaction: Gerischer model

● Global reaction:

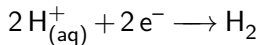


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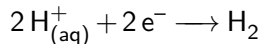


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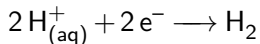


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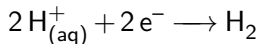


Motivation



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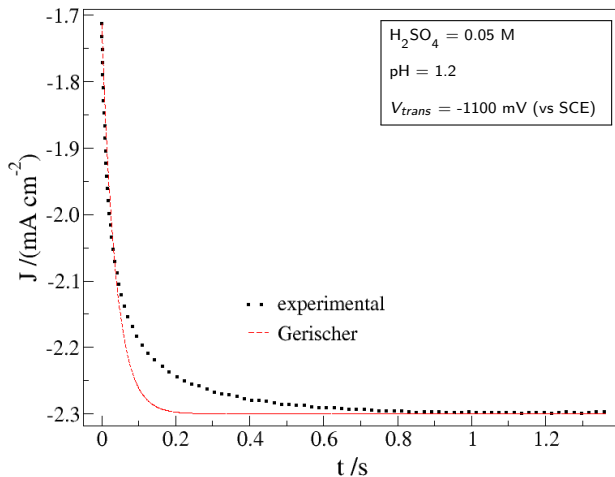
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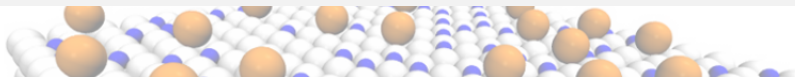
On Ag, HER occurs through a Volmer-Heyrovsky mechanism, being Volmer reaction the rate-determining step.

Motivation

Example of Gerischer model on Ag(100)



Methodology



Objectives

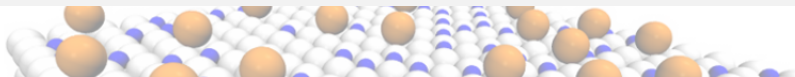
- To improve Gerischer's model by introducing several processes not taken into account
- To study the kinetics of HER on silver electrodes



General strategy

- To use KMC and DFT calculations to simulate the experimental results
- To get an analytical expression of the kinetic equations (?)
- To design new relevant experiments

Methodology



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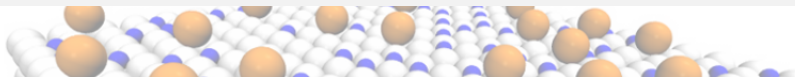
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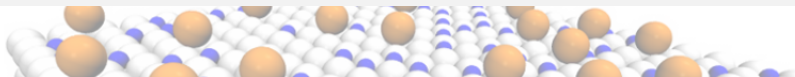
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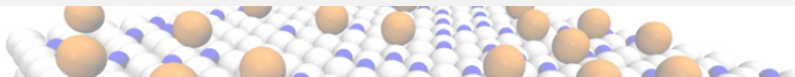
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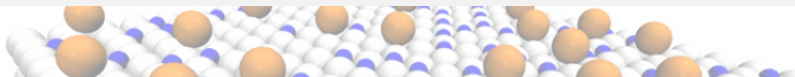
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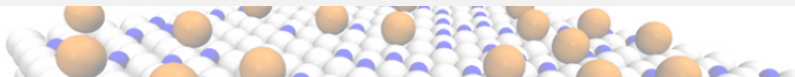
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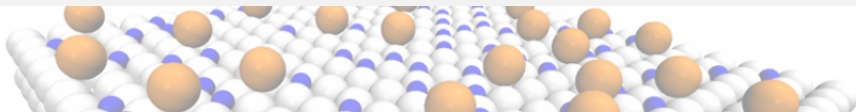
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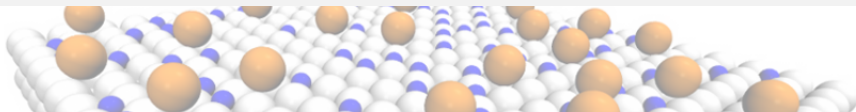


Processes included in KMC calculations

- Repulsion between H atoms ●
- H surface diffusion ●
- Tafel reaction
- Partial loss of active sites
- H absorption
- Desorption of anions ●



Methodology

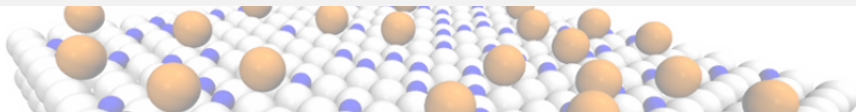


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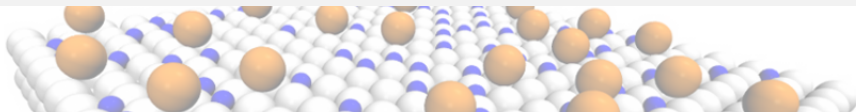


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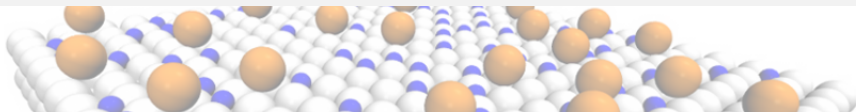


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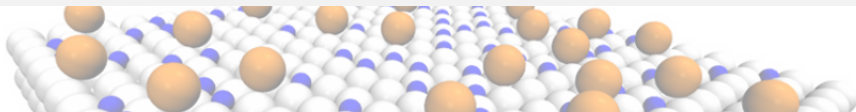


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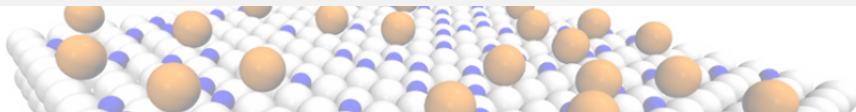


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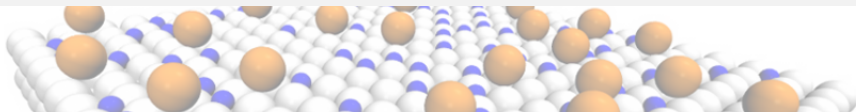


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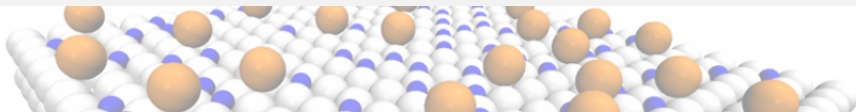


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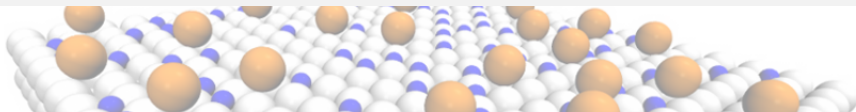


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Methodology



Processes included in KMC calculations

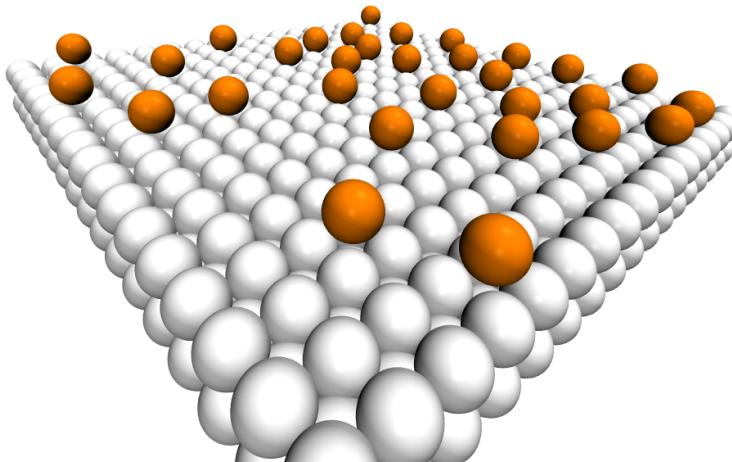
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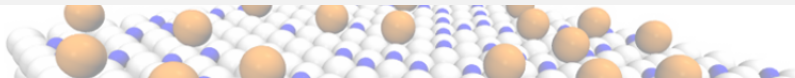
Methodology



A KMC movie



Methodology

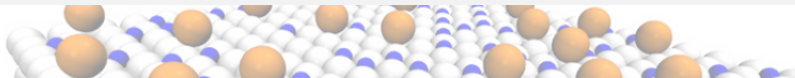


● HER reaction: modified Gerischer model

$$j_v = k_v(1 - \theta_H - \theta_A) \quad (1) \text{ Volmer current}$$



Methodology



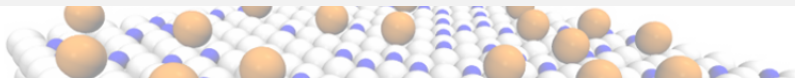
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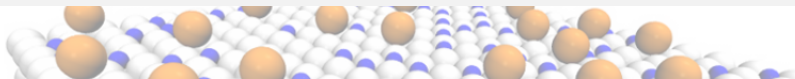
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Methodology



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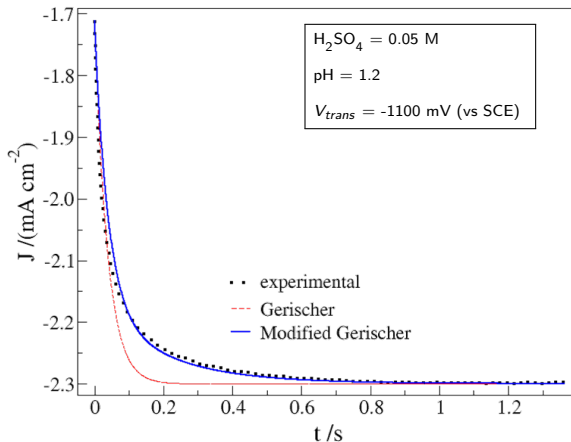
θ_A : concentration of anions

- Anions distribution is homogeneous.
- There are no lateral interactions between the anions.

Methodology

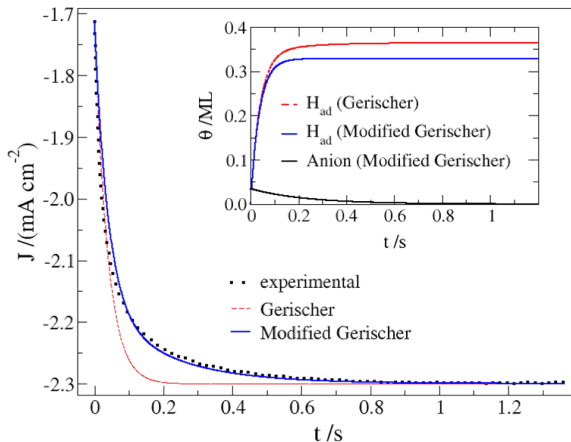


Applying modified Gerischer model on Ag(100)



Methodology

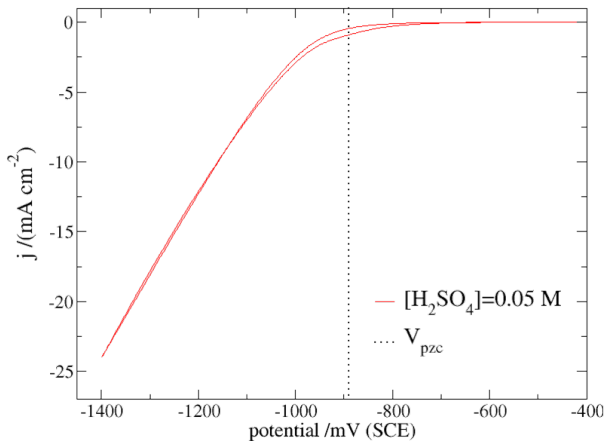
Applying modified Gerischer model on Ag(100)



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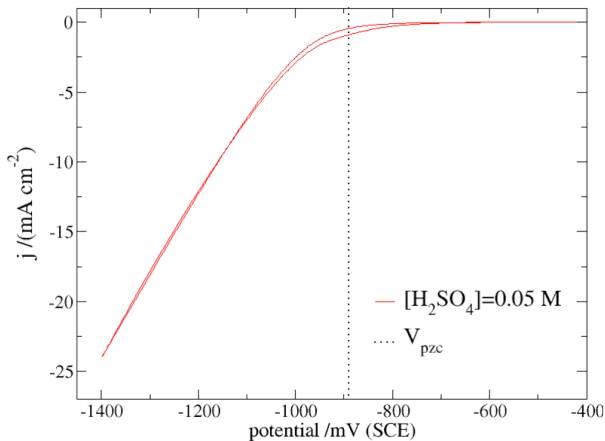
Experimental setup



Methodology



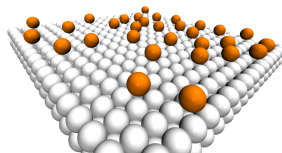
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Methodology



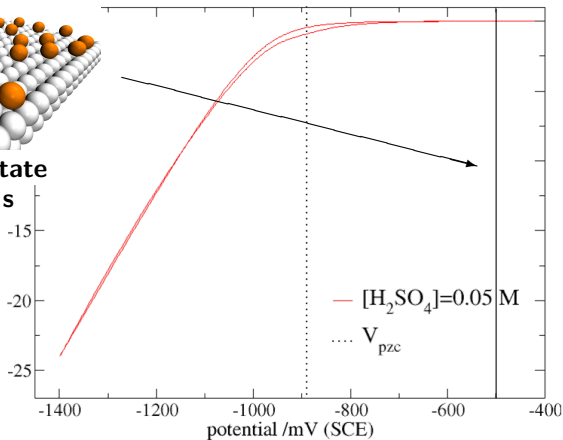
Experimental setup



Initial state

$t < 0\text{ s}$

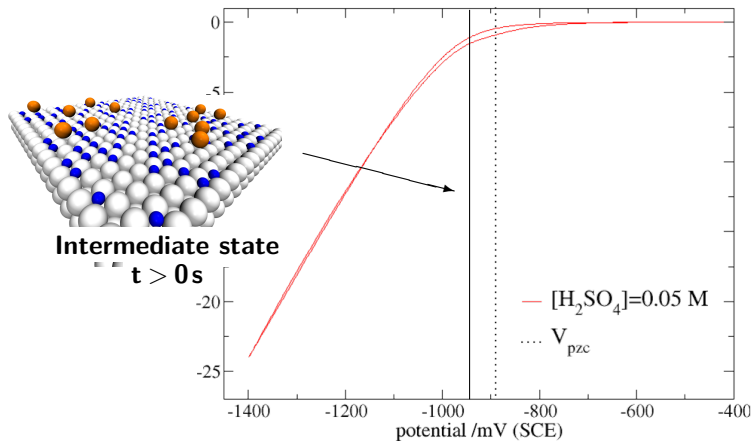
$j/(\text{mA})$



Methodology

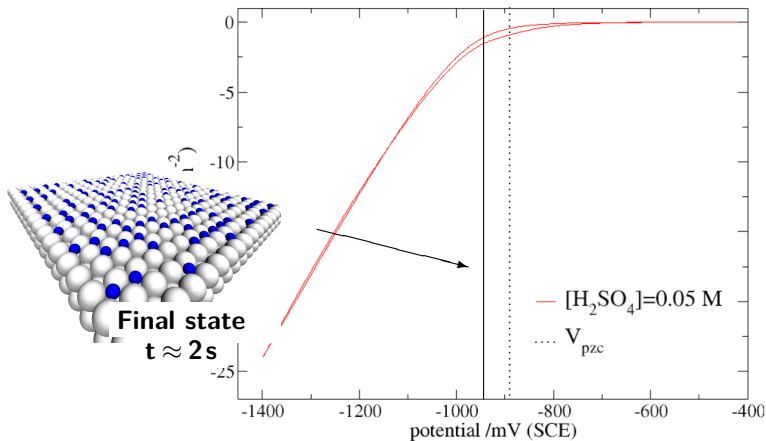


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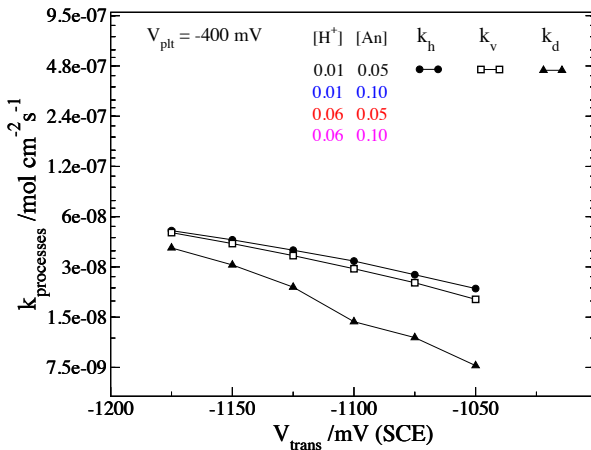
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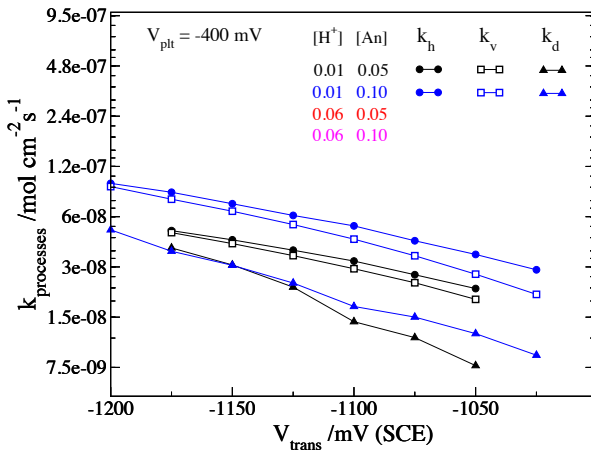
Results

Kinetic constants



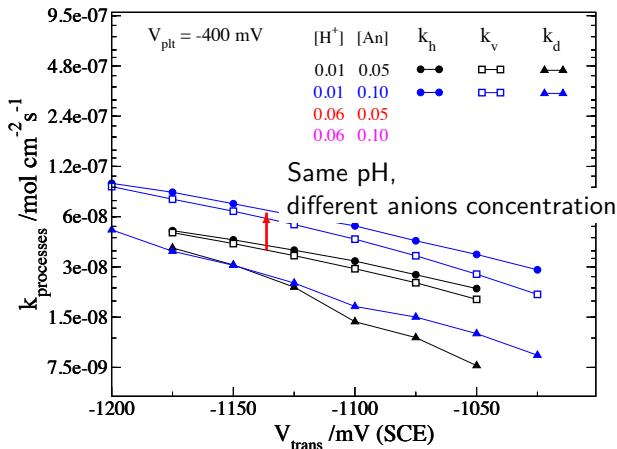
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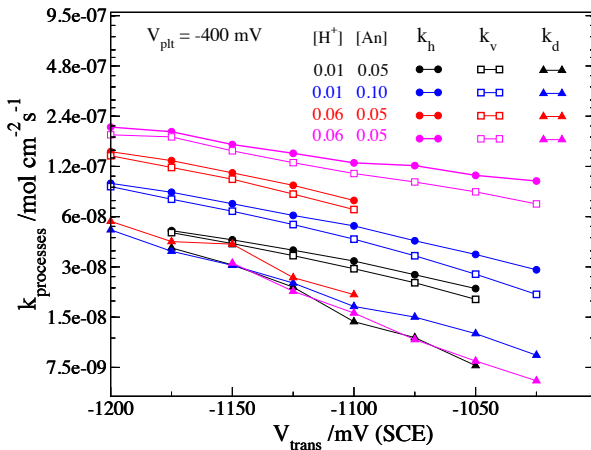
Results

Kinetic constants ●



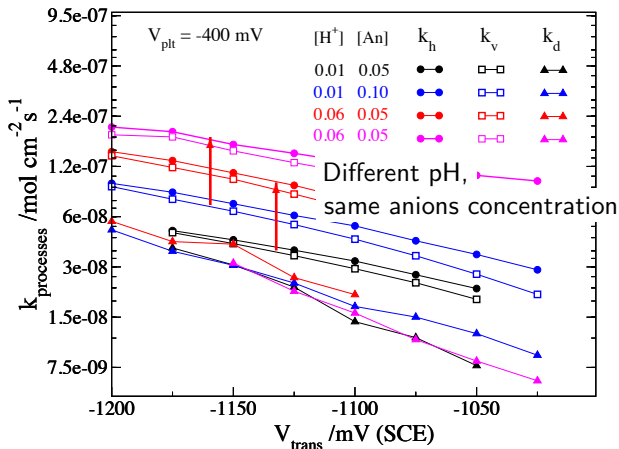
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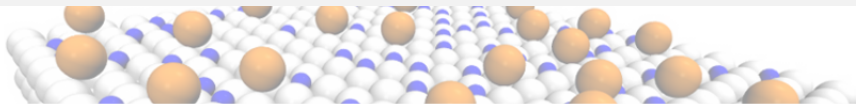


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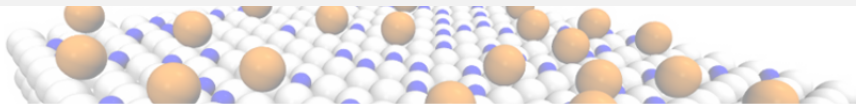
Conclusions and Perspectives



Conclusions

- Anion desorption is an important reaction simultaneously occurring during HER
- Including anion desorption improves Gerischer model for HER on silver
- Anions adsorbed in the surface have a catalytic effect on the reaction
- Even a small concentration of anions has a considerable effect on the potentiostatic response

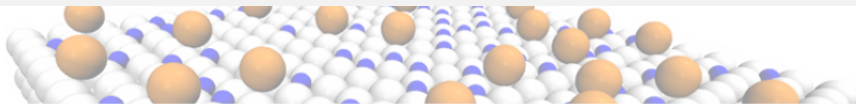
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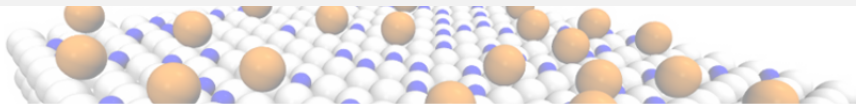
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